Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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: Page 1 of ??

Assignment 1

COT 6410 : Spring 2325 : Dr. Vassbinder
Due Thursday, February 26, 2325

Daftar Isi

Problem 1 **3**
(a) . . . 3
(b) . . . 3

Problem 2 **3**
(a) . . . 3
(b) . . . 4

Problem 3 **5**

Problem 1

Problem 1Problem 1 continued on next page... Problem 1 (continued)Problem 1 continued on next page... Define ND or $NotDominating = \{\ f \mid forsomex, f(x) < x\ \}$.

(a)

Problem 1 [(a)]Problem 1 [(a)] continued on next page... Problem 1 [(a)] (continued)Problem 1 [(a)] continued on next page... Show some minimal quantification of some known primitive recursive predicate that provides an upper bound for the complexity of ND.
[breakable,enhanced,colback=gray!5!white,title=Answer] We can quantify with primitive recursive predicates like so:
$\exists\langle x,t\rangle[STP(f,x,t)\ \&\&\ VALUE(f,x,t) < x]$

Because of the quantifier $\exists$, the problem is no harder than a problem in the RE (or Recursively Enumerable) problem class. Problem 1Problem 1 continued on next page... Problem 1 (continued)Problem 1 continued on next page...

(b)

Problem 1 [(b)]Problem 1 [(b)] continued on next page... Problem 1 [(b)] (continued)Problem 1 [(b)] continued on next page... Show that $K \leq_m ND$, where $K = \{\ f \mid f(f)converges\ \} = \{\ f \mid \varphi_f(f)\downarrow\ \}$.
[breakable,enhanced,colback=gray!5!white,title=Answer] Let f be an arbitrary index of function.
Define $g_f(z) = \varphi_f(f) - \varphi_f(f) + z$
$f \in K \forall z\ g_f(z) = zg_f(z)\downarrow g_f \in ND$
$f \notin K \forall z\ g_f(z)\uparrow g_f \notin ND$
Thus, $f \in K \iff g_f \in ND$
And so, $K \leq_m ND$ Problem 1Problem 1 continued on next page... Problem 1 (continued)Problem 1 continued on next page... Problem 1 (continued)Problem 1 continued on next page... Problem 1

Problem 2

Problem 2Problem 2 continued on next page... Problem 2 (continued)Problem 2 continued on next page... Consider the languages:

$$L = \{\ a^m b^n c^t \mid t = min(m,n)\ \}$$

(a)

Problem 2 [(a)]Problem 2 [(a)] continued on next page... Problem 2 [(a)] (continued)Problem 2 [(a)] continued on next page... Use the **Myhill-Nerode Theorem** to show that L **is not** a Regular Language.

[breakable,enhanced,colback=gray!5!white,title=Answer] **Proof Idea**

We need to find the equivalence classes of the language L, and show that you would need an infinite number of states to implement them. As an FSA cannot have infinite states, L cannot be regular.

Proof
$a^i b^j c^j \notin L$, where $j > i$
$a^i b^j c^j \in L$, where $i \geq j$
Because there are no limits on i, there are a countably infinite number of these classes. Because a FSA must have a finite number of states, you cannot construct one to recognize L. Therefore L is not a Regular Language. Problem 2Problem 2 continued on next page... Problem 2 (continued)Problem 2 continued on next page...

(b)

Problem 2 [(b)]Problem 2 [(b)] continued on next page... Problem 2 [(b)] (continued)Problem 2 [(b)] continued on next page... Use the **Pumping Lemma for CFLs** to show that L **is not** a Context Free Language.

[breakable,enhanced,colback=gray!5!white,title=Answer] This is a proof by contradiction.
Assume L is a Context Free Language.
And the Pumping Lemma for CFL states that any string $s \in L = uvxyz$, where:
1. $\forall i \geq 0,\ uv^i xy^i z \in L$
2. $\mid vy \mid > 0$
3. $\mid vxy \mid \leq p$
There are 2 cases to consider: (1) $t = n$, and (2) $t = m$.

Case 1
Choose string $s = a^m b^n c^t$, where $t = n$.
Then there are more a than b, for example $a^3 b^2 c^2 = aaabbcc$.
Let i be 0.
If vxy contains both a and c, then with $i = 0$, the number of c are not equal to either a or b.
(for example $a^3 b^2 c^2 = aaabbcc = aaa^0 bbc^0 c = aabbc \notin L$)
Thus we have a contradiction.

Case 2
Choose string $s = a^m b^n c^t$, where $t = m$.
Then there are less a than b, for example $a^2 b^3 c^2 = aabbbcc$.
Let i be 0.
If vxy contains a and b but no c, then with $i = 0$, $t \neq m$ and the number of $c \neq \min(m, n)$.
(for example $a^2 b^3 c^2 = aabbbcc = aa^0 bbb^0 cc = abbcc \notin L$)
Thus we have a contradiction.

Therefore L cannot be a Context Free Language. Problem 2Problem 2 continued on next page... Problem 2 (continued)Problem 2 continued on next page... Problem 2 (continued)Problem 2 continued on next page... Problem 2

Problem 3

Problem 3Problem 3 continued on next page. . . Problem 3 (continued)Problem 3 continued on next page. . . Explain what a tachyon is.

[breakable,enhanced,colback=gray!5!white,title=Answer] A tachyon is a subatomic particle that naturally exists at faster-than-light velocities. They can often be associated with time travel or produced as a byproduct of temporal distortions. Tachyons can exist both naturally, as in the tachyon eddies in the Bajoran system, and as the byproducts of certain technologies, such as cloaking devices and transporters. The detection of tachyons may lead to clues about recent temporal or spactial anomalous. Problem 3 (continued)Problem 3 continued on next page. . . Problem 3